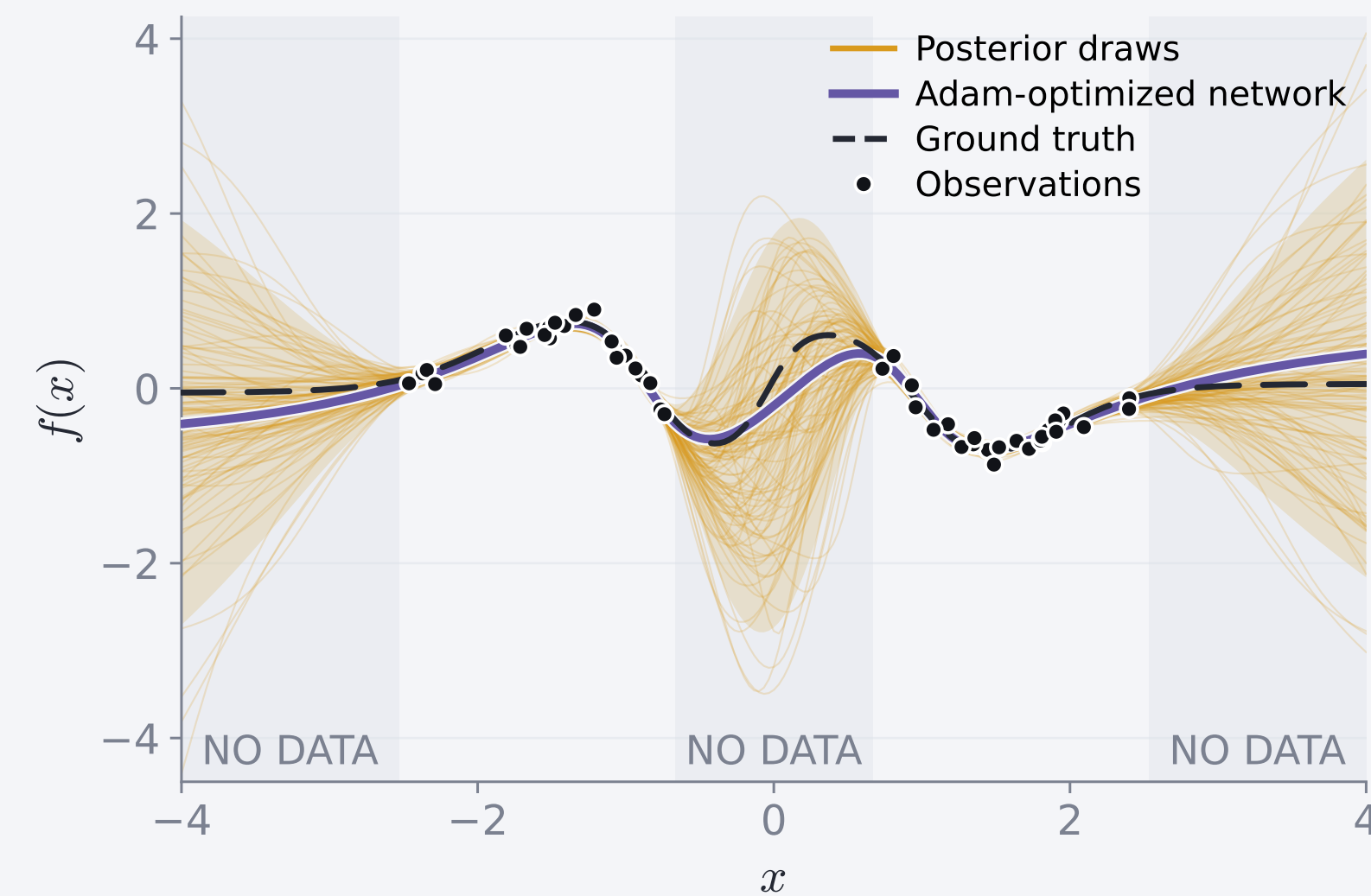




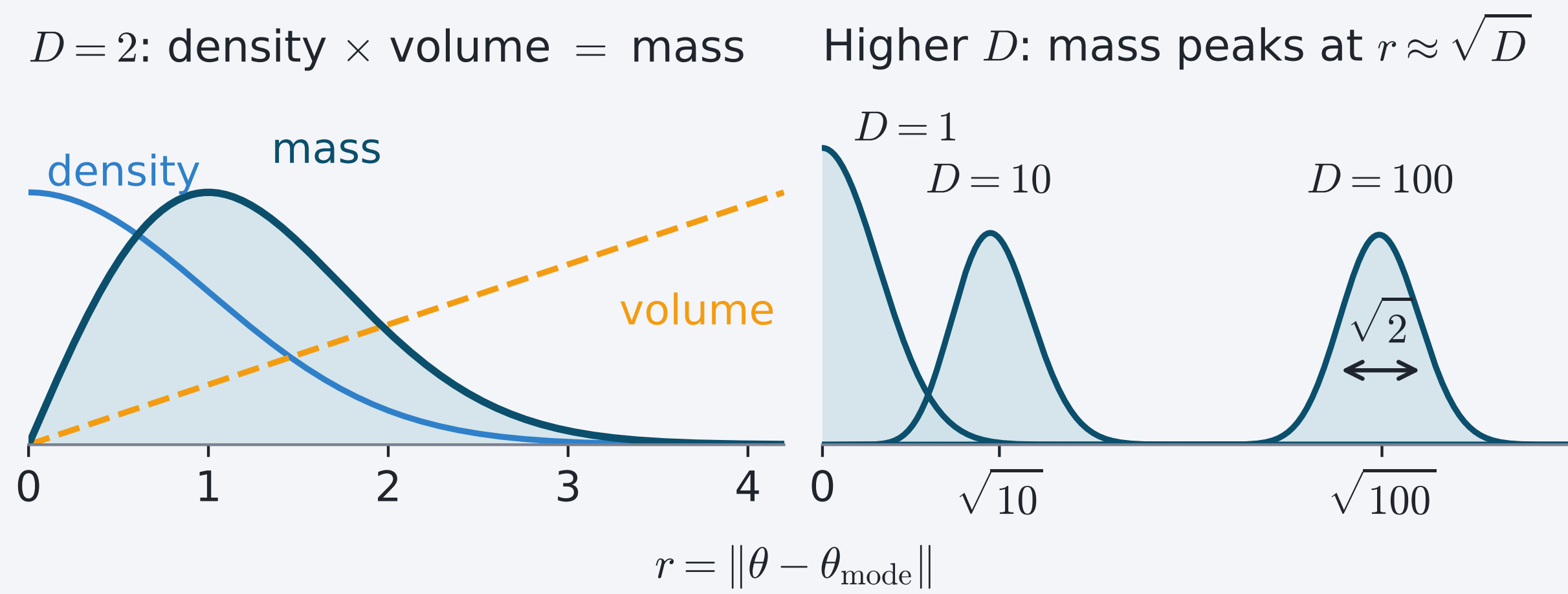
## A trained network is a single guess



Ordinary training gives one set of weights, one function. Bayesian sampling gives a distribution over weights  $\theta$ , so uncertainty over functions.

$$\pi(\theta) = p(\theta | \mathcal{D}) \propto \underbrace{p(\mathcal{D} | \theta)}_{\text{fit}} \underbrace{p(\theta)}_{\text{prior}}$$

## Probability lives on a thin shell



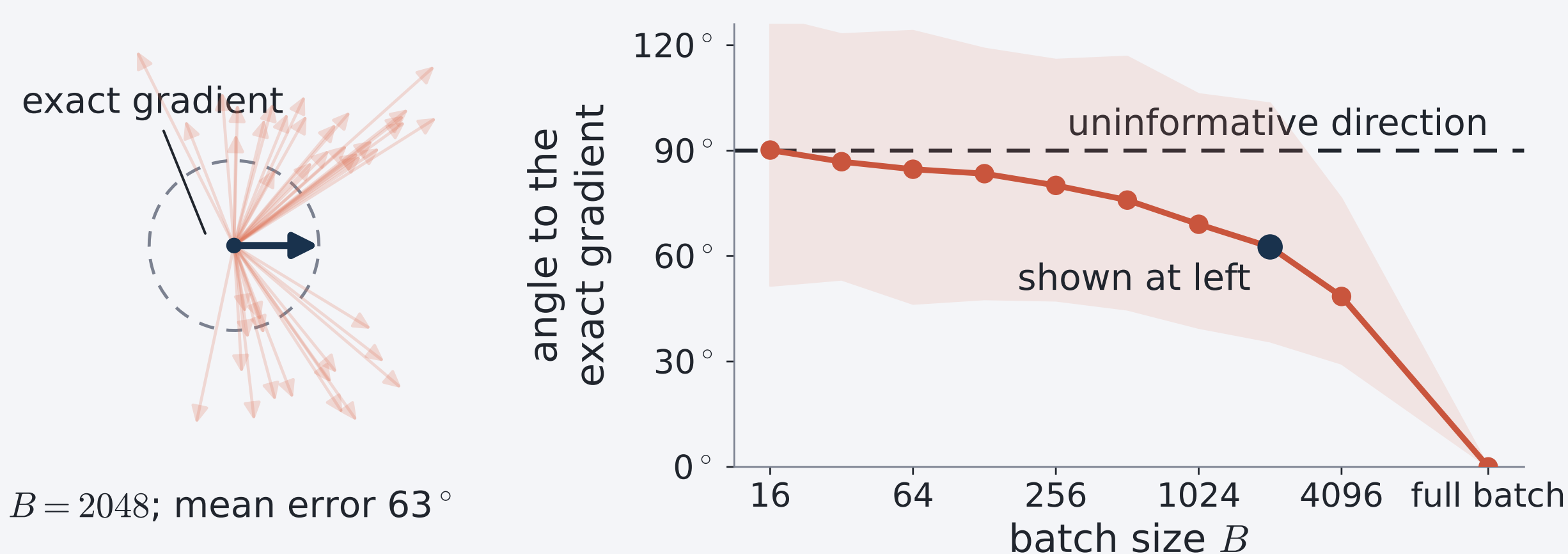
Because volume grows as  $r^{D-1}$ , probability concentrates in the *typical set*: a thin shell near  $\|\theta\|^2 \approx D$ . A correct sampler therefore spends almost all its time on this shell.

## Minibatch gradients are cheap but noisy

Markov chain Monte Carlo (MCMC) generates a sequence of correlated samples targeting  $\pi$ . Gradient-based samplers use  $\nabla \log \pi(\theta)$  to guide their updates. Computing this gradient exactly uses all  $N$  data points. A minibatch of size  $B \ll N$  is unbiased and roughly  $N/B$  times cheaper, but never exact:

$$\hat{g}_B(\theta) = \nabla \log p(\theta) + \frac{1}{B} \sum_{i \in B} \nabla \log p(\mathcal{D}_i | \theta), \quad \text{Cov}[\hat{g}_B] \propto 1/B.$$

Example: UCI Superconductivity Bayesian linear regression ( $D = 81$ ,  $N = 17,010$ ).



## Noise heats HMC but only turns a ray

Hamiltonian Monte Carlo (HMC)<sup>[1]</sup> and ray tracing (RT)<sup>[2]</sup> are gradient-based samplers that both use  $\nabla \log \pi$ . HMC evolves an unconstrained velocity, whereas RT evolves a unit direction.

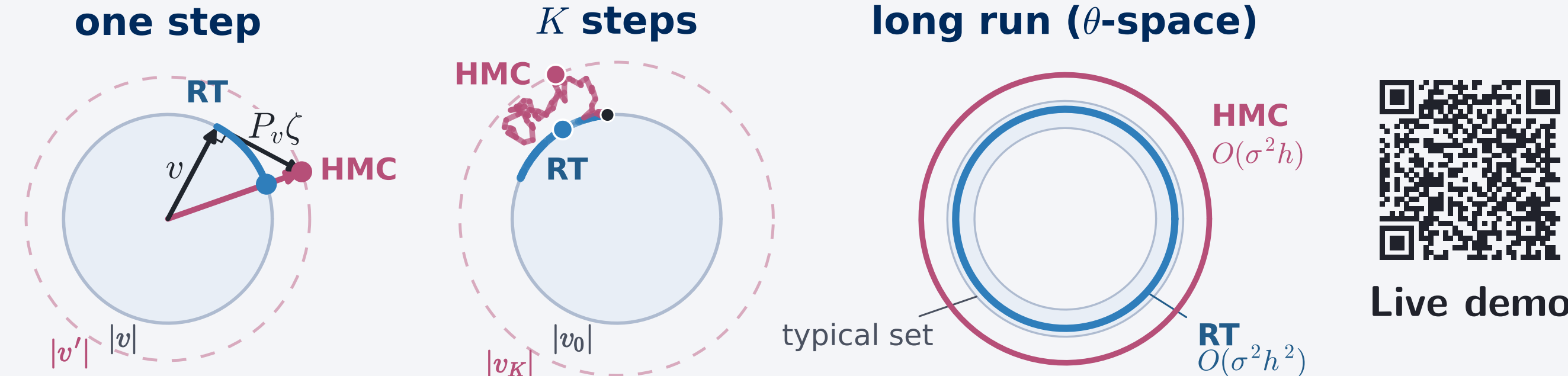
### 1. One noisy step

Without a Metropolis correction, mean-zero gradient noise raises HMC's expected kinetic energy, increasing its speed.

For RT, write the isotropic gradient error as  $\sigma\zeta$ . With  $n(\theta) = \pi(\theta)^{1/(D-1)}$ , the unit direction obeys

$$\frac{du}{ds} = P_u \left( \nabla \log n(\theta) + \frac{\sigma}{D-1} \zeta \right), \quad P_u = I - uu^\top.$$

$P_u$  projects onto the tangent plane, so noise can rotate  $u$  but cannot change its norm.



### 2. Along a trajectory

With step size  $h$  and trajectory length  $S$ , the  $S/h$  noisy kicks accumulate as

$$\frac{S}{h} \cdot O(\sigma^2 h^2) = O(\sigma^2 h).$$

Let  $\sigma_c(h)$  be the largest noise scale at step  $h$  that keeps long-run bias below a fixed tolerance, with  $\sigma_c(h) \propto h^{-p}$ . For unadjusted minibatch HMC,  $p = \frac{1}{2}$ .

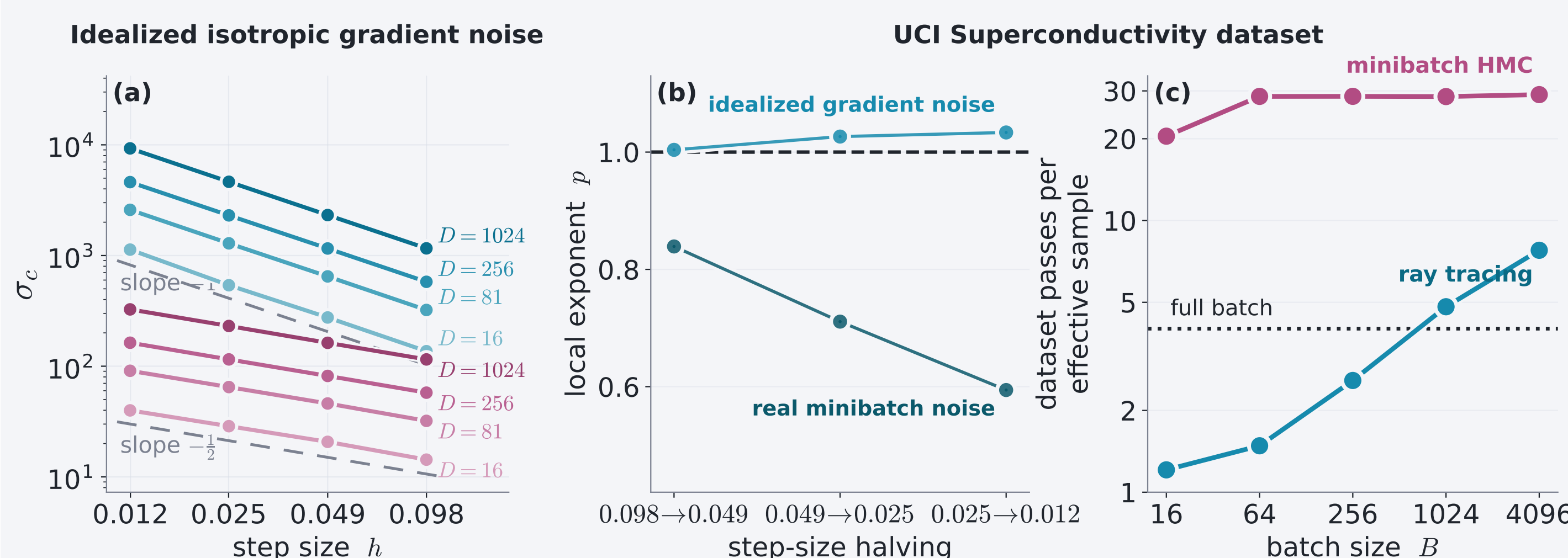
### 3. Long-run bias

For RT, isotropic rotations leave the uniform direction distribution unchanged, so the  $O(\sigma^2 h)$  effect creates no leading bias:

$$b_{\text{RT}}(h, \sigma) = O(\sigma^2 h^2) + o(\sigma^2) \Rightarrow \sigma_c(h) = \Omega(h^{-1}), \quad p \geq 1$$

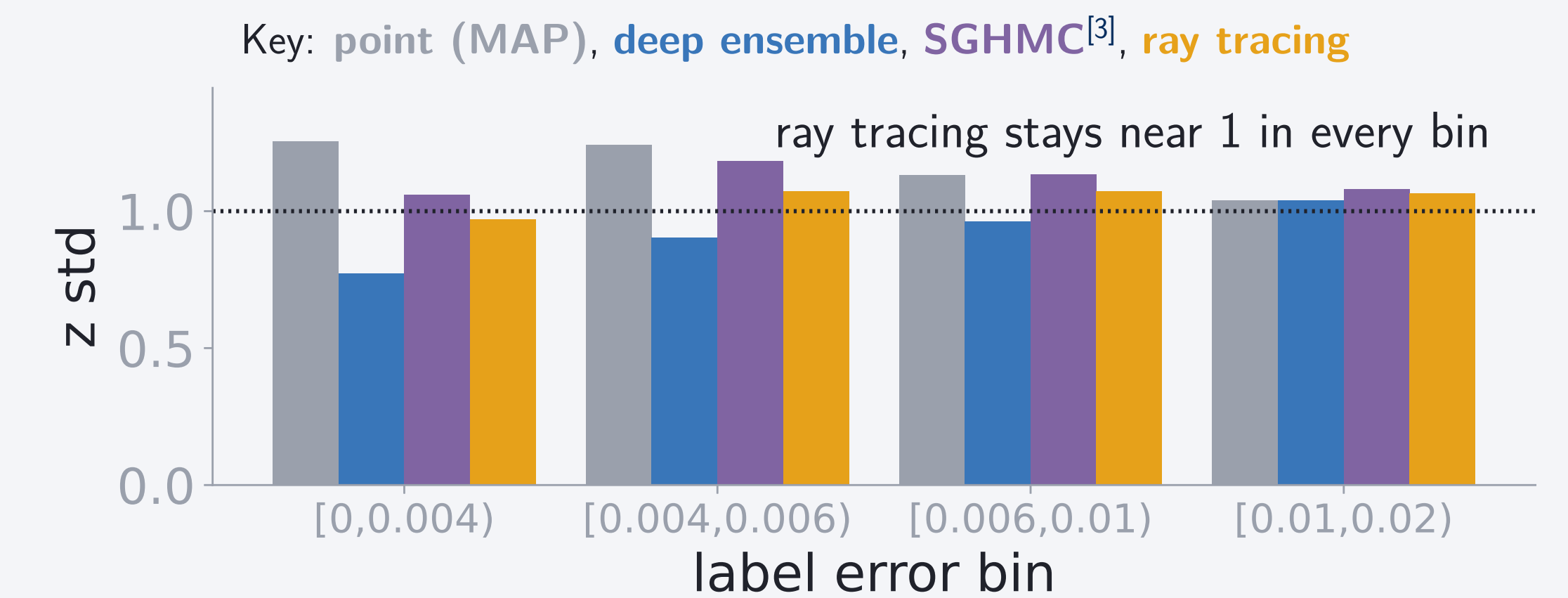
## Measured noise tolerance and sampling cost

Across  $D = 16$ – $1024$ ,  $\sigma_c \propto h^{-1}$  for RT and  $\sigma_c \propto h^{-1/2}$  for HMC; real minibatch noise weakens RT's scaling.

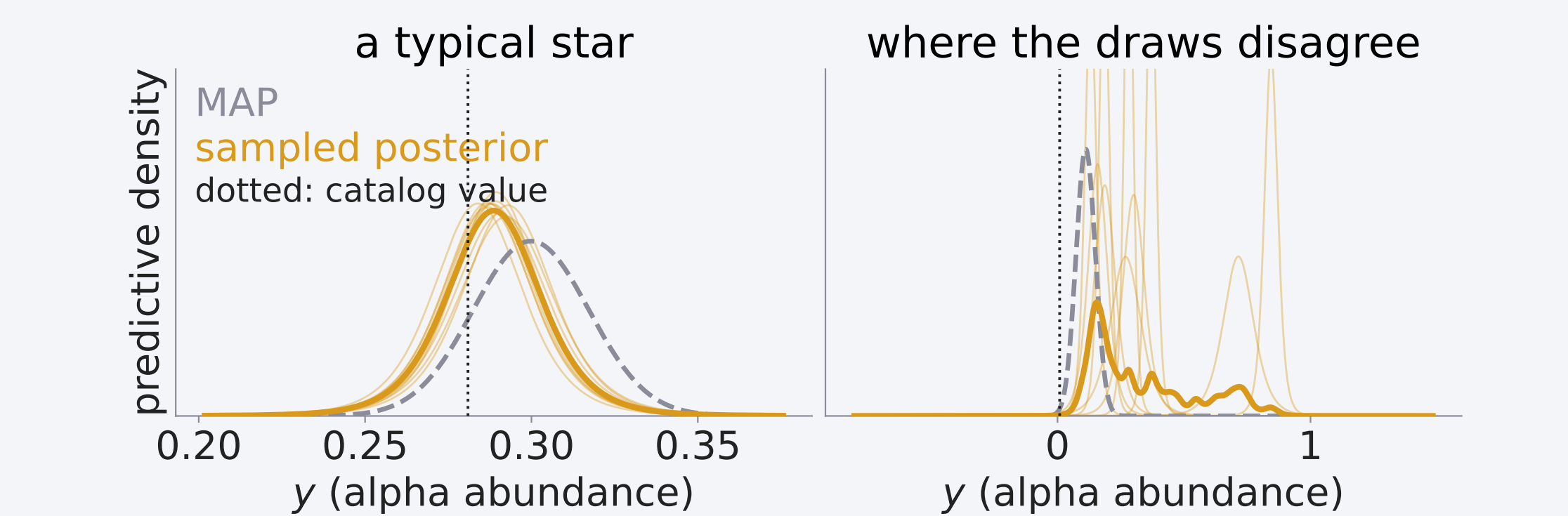
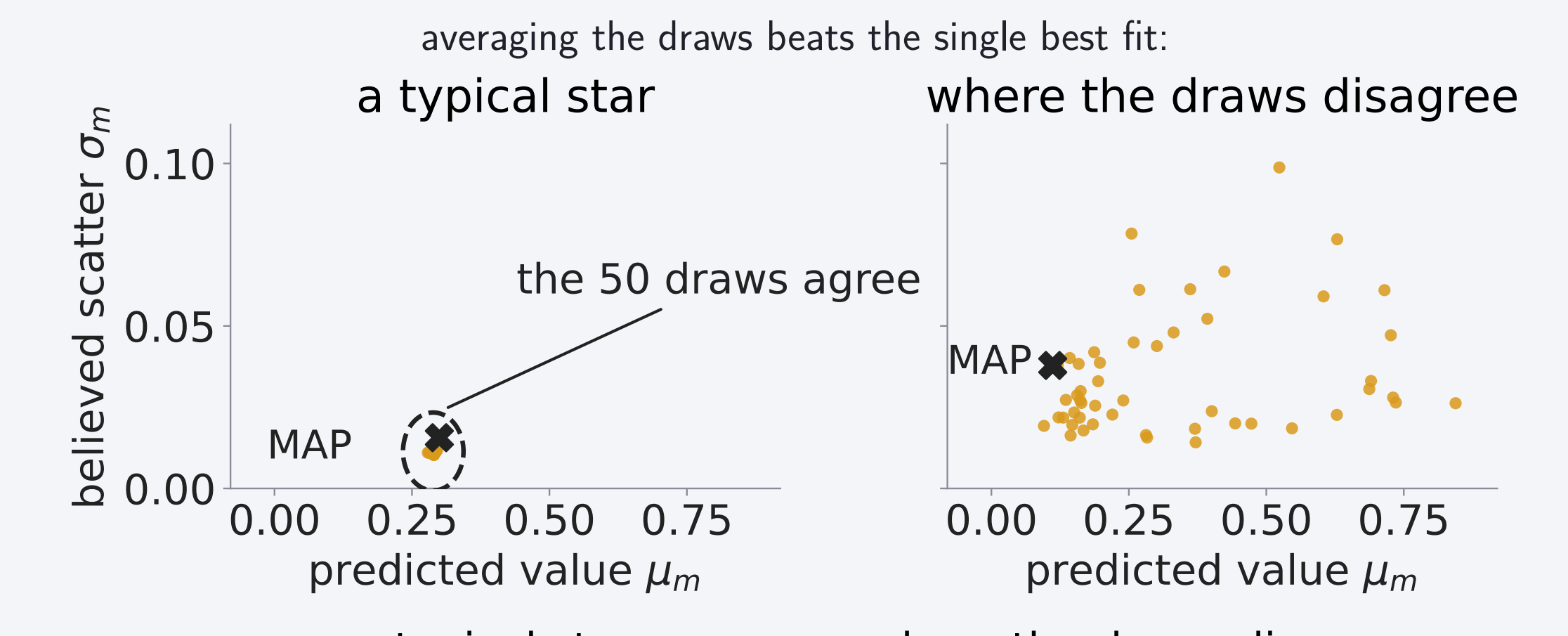


## Empirical study on real Gaia data

We predict each star's alpha abundance plus a per-star scatter from its Gaia XP spectrum (110 coefficients), for 126,156 giants with APOGEE labels<sup>[4,5]</sup>, using a  $D = 11,394$  network sampled with minibatch ray tracing.



z std = actual / claimed error: closer to 1 is better.



## Conclusion

- We took the sampler from its paper to a real survey posterior. The draws are the product: averaging them beat the single best fit, and their spread warns us when the model is guessing.
- The gains held on real data: held-out stars were 27% more probable than under the single fit, and an effective sample took  $3.3\times$  less data than full-batch sampling.
- The theory held as well: gradient noise can only turn a constant-speed ray, and the predicted noise tolerance ( $p = 1$ , against  $p = \frac{1}{2}$  for HMC) matched at every dimension we tested.

### References

- [1] Betancourt, M. 2017, *A Conceptual Introduction to Hamiltonian Monte Carlo*, arXiv:1701.02434
- [2] Behroozi, P. 2026, *The Ray Tracing Sampler: Bayesian Sampling of Neural Networks for Everyone*, Open Journal of Astrophysics 9, doi:10.33232/001c.165932
- [3] Chen, T., Fox, E. B., Guestrin, C. 2014, *Stochastic Gradient Hamiltonian Monte Carlo*, ICML, PMLR 32(2), 1683
- [4] Gaia Collaboration, Vallenari, A., et al. 2023, *Gaia Data Release 3: Summary of the content and survey properties*, A&A 674, A1
- [5] Laroche, A., Speagle, J. S. 2025, *Closing the Stellar Labels Gap: Stellar Label Independent Evidence for  $[\alpha/M]$  Information in Gaia BP/RP Spectra*, ApJ 979, 5

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